Evolution in Audio Deepfake Detection

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  (202521004)
  Course evaluation presentation of **Advance Machine Learning (AML)**
  under Supervision of
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Presentation Overview

- Introduction to Audio Deepfake Deepfake (ADD) and literature survey.
- Introduction to RawNet2 -> AASIST -> AASIST3 -> Rawformer.
- Experimental setup & Results.
- Summary and conclusion.
- Open research problems.

What is Deepfake?

1. Any AI generated content, produced via deep learning method can be called as deepfake.
2. Deepfakes are generally created to fool humans rather than spoofing systems.
3. Applications:
   - Social media platforms
   - Voice-controlled assistants
   - Biometric authentication systems

RawNet2

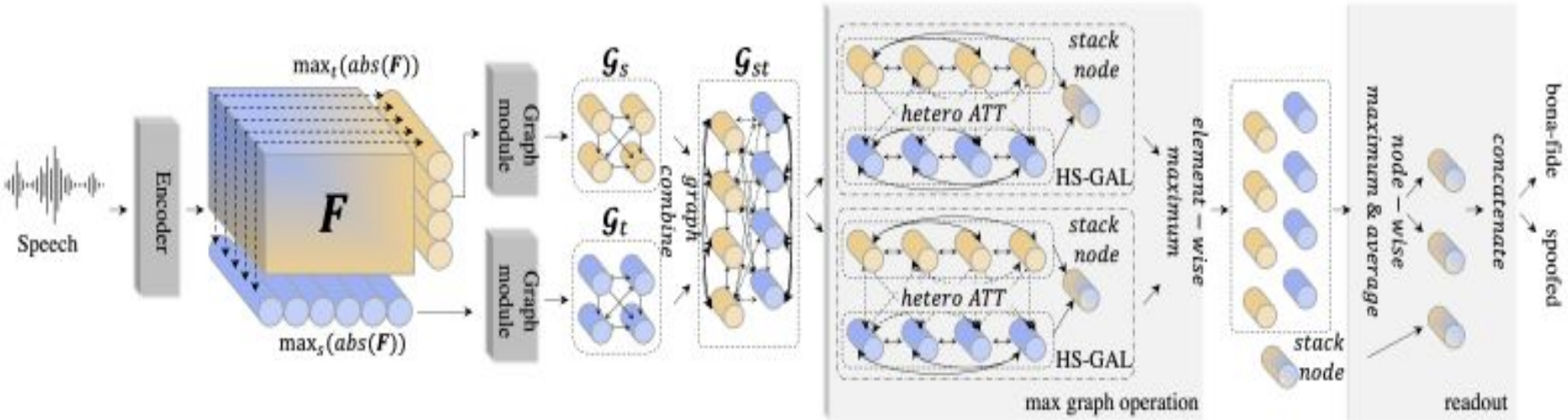
Layer	Input:**64000** samples		Output shape
Fixed Sinc filters	Conv(**129**,1,128) Maxpooling(3) BN & LeakyReLU		(**21290**,128)
Res block	BN & LeakyReLU Conv(3,1,128) BN & LeakyReLU Conv(3,1,128) Maxpooling(3) FMS	$\times$ 2	(**2365**,128)
Res block	BN & LeakyReLU Conv(3,1, **512**) BN & LeakyReLU Conv(3,1, **512**) Maxpooling(3) FMS	$\times$ 4	(**29**,512)
GRU	GRU(1024)		(1024)
FC	1024		(1024)
Output	**1024**		**2**

Source: Tak, H., Patino, J., Todisco, M., Nautsch, A., Evans, N., & Larcher, A. (2021, June). End-to-end anti-spoofing with rawnet2. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 6369-6373).

RawNet2 Sub-Modules Overview

- SincConv Frontend → Raw waveform → learnable band-pass filters (mel-init)
- Initial Processing → |x| + pooling + BN + SELU
- Residual CNN Stack
- Channel Attention & GRU layer
- FC Head → Final classification

AASIST Overview

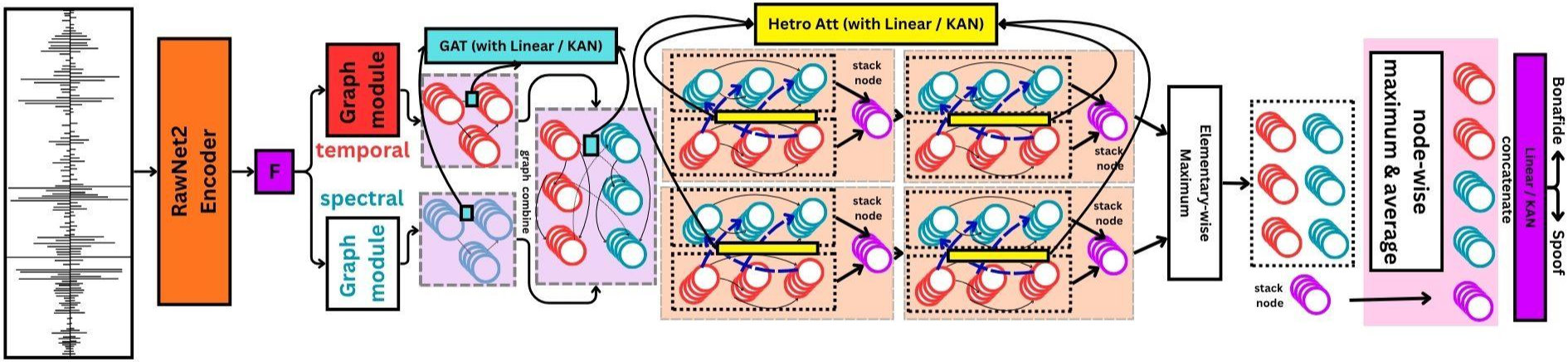


Source: Jung, Jee-weon, et al. "Aasist: Audio anti-spoofing using integrated spectro-temporal graph attention networks." *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, 2022.

AASIST Sub-Modules Overview

- RawNet2 Encoder
- Lower Feature Maps (LFM) & Higher Feature Maps (HFM).
- Graph Modules & Graph Combine
- Heterogeneous Attention & Stack node
- Element wise Max/Min operator & Readout

AASIST-3 Overview

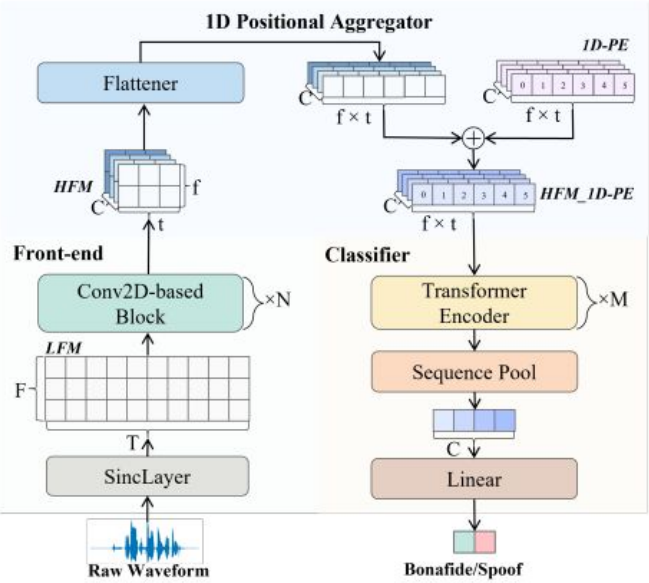


Source: Borodin, Kirill, et al. "AASIST3: KAN-enhanced AASIST speech deepfake detection using SSL features and additional regularization for the ASVspoof 2024 Challenge." *Proc. ASVspoof, INTERSPEECH,* 2024.

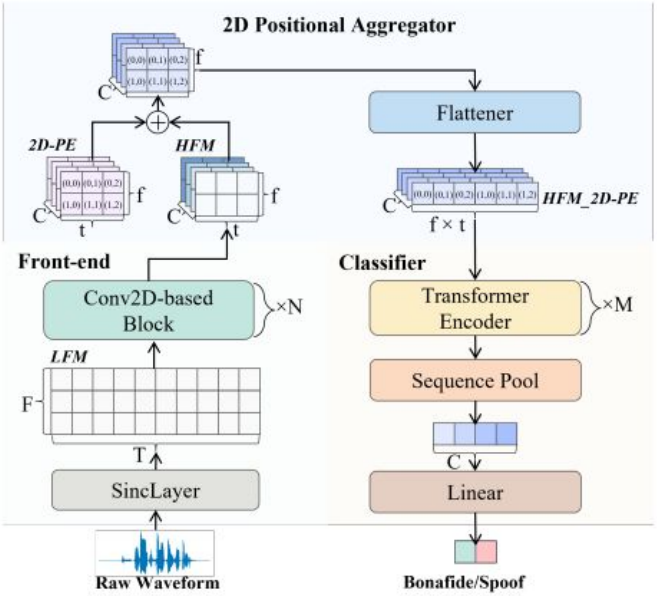
AASIST3 Sub-Modules Overview

- RawNet2 Encoder
- Lower Feature Maps (LFM) & Higher Feature Maps (HFM).
- Graph Modules (With KAN) & Graph Combine (With KAN)
- Heterogeneous Attention & Stack node
- Element wise Max/Min operator & Readout (Via KAN Linear layer)

RawFormer Overview



(a) Rawformer with 1D positional encoding.



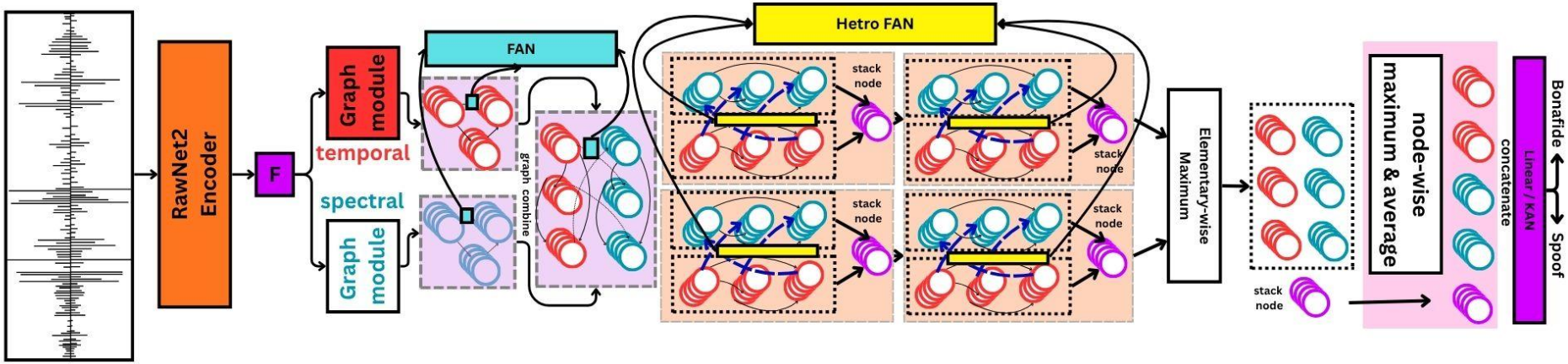
(b) Rawformer with 2D positional encoding.

Source: Liu, X., Liu, M., Wang, L., Lee, K. A., Zhang, H., & Dang, J. (2023, June). Leveraging positional-related local-global dependency for synthetic speech detection. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 1-5).

RawFormer Sub-Modules Overview

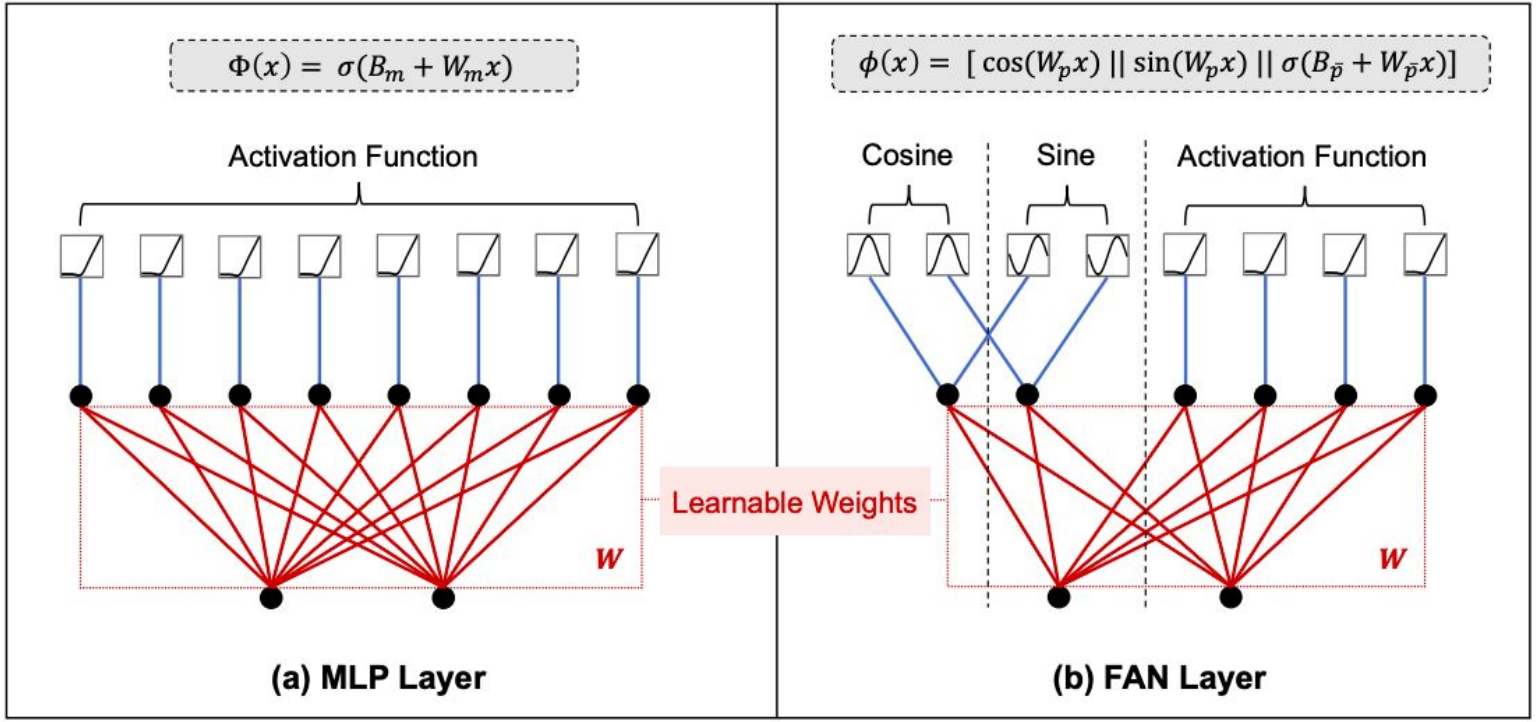
- Pre-Emphasis + Sinc Frontend
- 2D CNN Feature Extractor (LFM → HFM)
- Positional Encoding + Flattening
- Transformer Encoder Stack (MHSA + FFN)
- Sequence Pooling + Classification

FAN-AASIST Overview



Source: Liu, X., Liu, M., Wang, L., Lee, K. A., Zhang, H., & Dang, J. (2023, June). Leveraging positional-related local-global dependency for synthetic speech detection. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 1-5).

FAN-AASIST Overview



Source: Yihong Dong, Ge Li, Yongding Tao, Xue Jiang, Kechi Zhang, Jia Li, Jinliang Deng, Jing Su, Jun Zhang, and Jingjing Xu, “Fourier analysis network,” in NeurIPS, 2025, San Diego, CA.

FAN-AASIST Sub-Modules Overview

- All the submodules are same as AASIST, Just replaced attention with FAN to capture sequential characteristics.
- Unlike self-attention, which computesdense pairwise interactions, FAN decomposes each input into periodic and non-periodic components.

FAN layer

$$\phi(x) = [\cos(W_p x) || \sin(W_p x) || \sigma(B_{\bar{p}} + W_{\bar{p}} x)]$$

Experimental Setup

Dataset: ASVSpoof 2019

Training duration of audios: 2 Seconds with 16k sampling rate

# of filters in Sinc-conv layer: 70

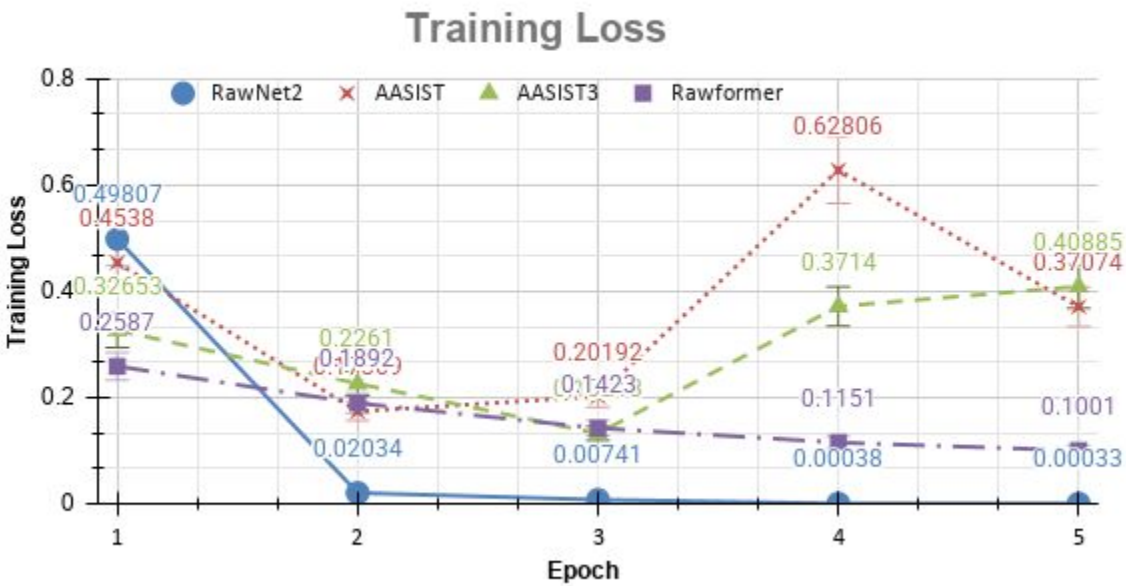
# of ResNet2 blocks in encoder: 06

Optimizer and Learning Rate: Adam, and 10^-4.

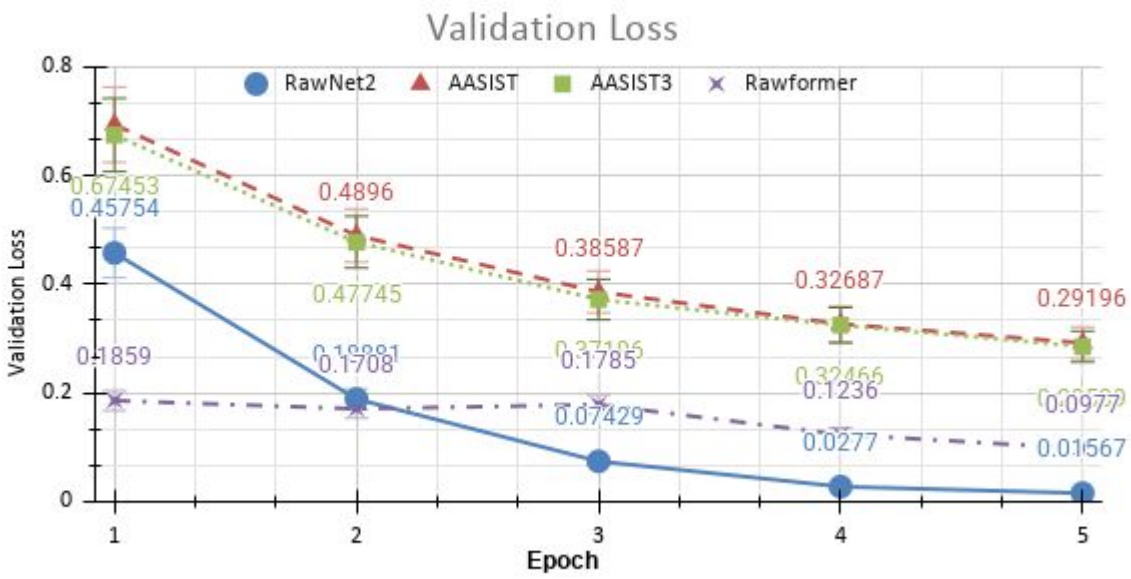
Epochs: 5

Performance Metrices: Variable

Experimental Results



Experimental Results



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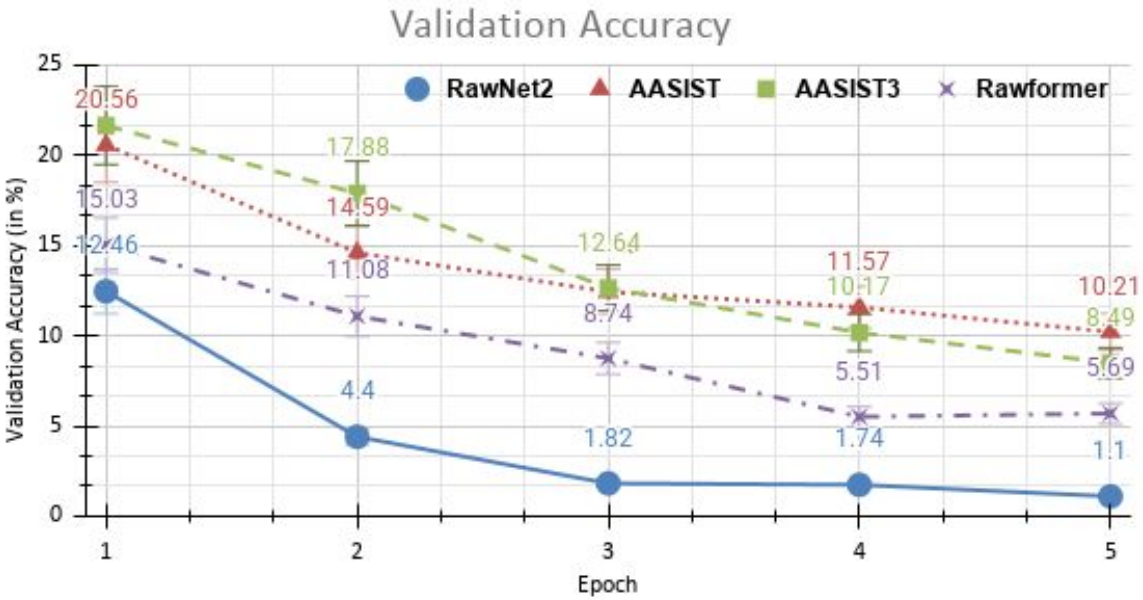

Experimental Results

Model	F1 Macro	F1 Micro	Precision Macro	Precision Micro	Recall Macro	Recall Micro	Hamming Loss	Balanced Accuracy	AUC	EER
RawNet2	0.4482	0.494	0.5841	0.494	0.7164	0.494	0.506	0.7164	0.8732	0.2214
AASIST	0.5553	0.6466	0.6031	0.6466	0.775	0.6466	0.3534	0.775	0.9081	0.1821
AASIST3	0.562	0.6542	0.6065	0.6542	0.7833	0.6542	0.3458	0.7833	0.9247	0.164
Rawformer	0.7369	0.8592	0.6971	0.8592	0.8592	0.8592	0.1408	0.8592	0.9426	0.1427

Model	Trainable Params	Total Params	Model Size (MB)
RawNet2	12,838,402	32,737,540	48.97
AASIST	297,866	1,312,218	1.14
AASIST3	1,451,311	6,004,919	5.54
Rawformer	345,132	2,234,021	1.32

Model	Input Shape	MACs (G)	FLOPs (G)	GFLOPs/sec
RawNet2	[1, 32000]	4.538	9.077	4.538
AASIST	[1, 32000]	4.754	9.508	4.754
AASIST3	[1, 32000]	4.827	9.653	4.827
Rawformer	[1, 32000]	6.186	12.371	3.093

Experimental Results

Model	Input (MB)	Forward/Backward (MB)	Params (MB)	Total (MB)
RawNet2	0.13	87.69	51.35	139.17
AASIST	0.13	434.61	1.18	435.92
AASIST3	0.13	434.88	0.85	435.85
Rawformer	0.13	322.39	0.69	323.21

Time taken to run codes

1. RawNet2 - 29 min 12 sec
2. AASIST - 55 min 32 sec
3. AASIST3 - 1 hr 43 mins
4. Rawformer - 1 hr 10 mins

Experimental Results (FAN)

```
Testing Accuracy: 0.7541165405617867
F1 Macro: 0.6147
F1 Micro: 0.7541
Precision Macro: 0.6101
Precision Micro: 0.7541
Recall Macro: 0.7473
Recall Micro: 0.7541
Hamming Loss: 0.2459
Balanced Accuracy: 0.7473
AUC: 0.8275
EER: 0.2515
```

```
==============================================================
MODEL PARAMETER SUMMARY
==============================================================
Trainable params          : 136,116
Total params              : 136,116
Model size (MiB)          : 0.52
==============================================================
```

```
==============================================================
FLOPs / MACs (per forward pass)
==============================================================
Input shape               : [1, 32000]
MACs                      : 4.548 G
FLOPs                     : 9.095 G
==============================================================
```

Observations: Not working good.

Summary & Conclusions

- I explored 4 existing models, and tried evolving it in one another model.
- I failed, but i learned
- Models and data organization.
- What is the key in evolution.
- Why Graph attention.
- Why sequential attention.
- What Can be next ??

=> Nested Learning

Refrences

[1] **ASV19 dataset =>** M. Todisco, X. Wang, V. Vestman et al., “Asvspoof 2019: Future horizons in spoofed and fake audio detection,” in Proc. INTERSPEECH, 2019.

[2] **RawNet2 =>** H. Tak, J. Patino, M. Todisco et al., “End-to-end anti-spoofing with rawnet2,” in Proc. ICASSP, 2021.

[3] **AASIST =>** Jung, Jee-weon, et al. "Aasist: Audio anti-spoofing using integrated spectro-temporal graph attention networks." *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, 2022.

[4] **AASIST3 =>** Borodin, Kirill, et al. "AASIST3: KAN-enhanced AASIST speech deepfake detection using SSL features and additional regularization for the ASVspoof 2024 Challenge." *Proc. ASVspoof, INTERSPEECH*, 2024.

[5] **Rawformer =>** Liu, X., Liu, M., Wang, L., Lee, K. A., Zhang, H., & Dang, J. (2023, June). Leveraging positional-related local-global dependency for synthetic speech detection. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 1-5).

[6] **FAN =>** Yihong Dong, Ge Li, Yongding Tao, Xue Jiang, Kechi Zhang, Jia Li, Jinliang Deng, Jing Su, Jun Zhang, and Jingjing Xu, “Fourier analysis network,” in NeurIPS, 2025, San Diego, CA.

Thank You …